physical processes, the laws satisfied by a physical system are not the same as those satisfied by the mirror reflection of that system. In fact, Feynman had a hand in the development of the physical theory which is able to accommodate this asymmetry. His account here is, accordingly, a dramatic one, as deeper and deeper mysteries of nature gradually unfold.

As physics develops, there are mathematical formalisms that develop with it, and which are needed in order to express the new physical laws. When the mathematical tools are skillfully tuned to their appropriate tasks, they can make the physics seem much simpler than otherwise. The ideas of vector calculus are a case in point. The vector calculus of three dimensions was originally developed to handle the physics of ordinary space, and it provides an invaluable piece of machinery for the expression of physical laws, such as those of Newton, where there is no physically preferred direction in space. To put this another way, the physical laws have a *symmetry* under ordinary rotations in space. Feynman brings home the power of the vector notation and the underlying ideas for expressing such laws.

Relativity theory, however, tells us that *time* should also be brought under the compass of these symmetry transformations, so a *four*-dimensional vector calculus is needed. This calculus is also introduced to us here by Feynman, as it provides the way of understanding how not only time and space must be considered as different aspects of the same four-dimensional structure, but the same is true of energy and momentum in the relativistic scheme.

The idea that the history of the universe should be viewed, physically, as a *four*-dimensional space-time, rather than as a three-dimensional space evolving with time is indeed fundamental to modern physics. It is an idea whose significance is not easy to grasp. Indeed, Einstein himself was not sympathetic to this idea when he first encountered it. The idea of space-time was not, in fact, Einstein's, although, in the popular imagination it is frequently attributed to him. It was the Russian/German geometer Hermann Minkowski, who had been a teacher of Einstein's at the Zurich